

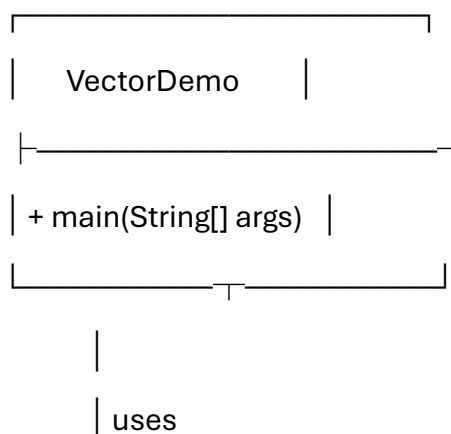
Aim

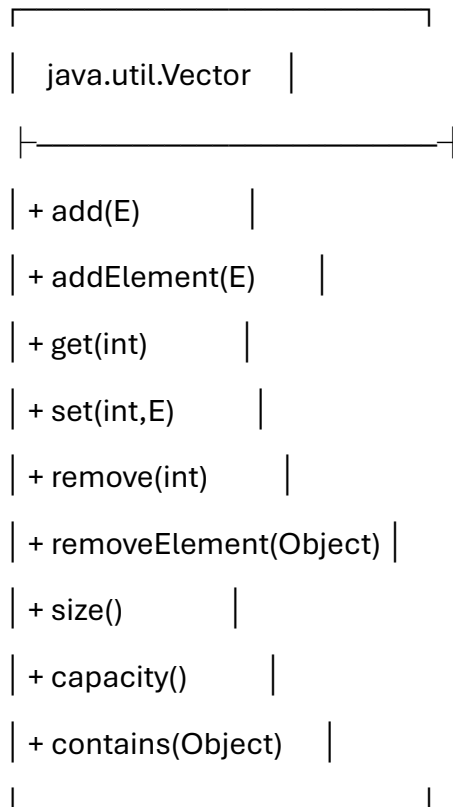
To write a Java program to demonstrate the important methods of the **Vector class**, such as `add()`, `addElement()`, `get()`, `set()`, `remove()`, `removeElement()`, `size()`, `capacity()`, and `contains()`.

Algorithm

1. Start the program.
2. Import the `java.util.Vector` class.
3. Create a `Vector` object.
4. Use `add()` to add elements to the `Vector`.
5. Use `addElement()` to add another element.
6. Display the `Vector`.
7. Use `get()` to retrieve an element at a specified index.
8. Use `set()` to replace an element.
9. Use `remove()` to remove an element using its index.
10. Use `removeElement()` to remove a specified element.
11. Use `size()` to find the number of elements.
12. Use `capacity()` to find the current capacity.
13. Use `contains()` to check whether an element exists.
14. Display all results.
15. Stop the program.

UML Diagram





Program:

```
package Observations;
```

```
import java.util.Vector;
```

```
public class VectorClass {
```

```
    public static void main(String[] args) {
```

```
        // Creating a Vector
```

```
        Vector<String> v = new Vector<>();
```

```
        // add()
```

```
        v.add("Apple");
```

```
v.add("Banana");
```

```
v.add("Mango");
```

```
System.out.println("After add(): " + v);
```

```
// addElement()
```

```
v.addElement("Orange");
```

```
System.out.println("After addElement(): " + v);
```

```
// get()
```

```
System.out.println("Element at index 1: " + v.get(1));
```

```
// set()
```

```
v.set(1, "Grapes");
```

```
System.out.println("After set(): " + v);
```

```
// remove(index)
```

```
v.remove(2);
```

```
System.out.println("After remove(2): " + v);
```

```
// removeElement()
```

```
v.removeElement("Orange");
```

```
System.out.println("After removeElement(): " + v);
```

```
// size()
```

```
System.out.println("Size of Vector: " + v.size());
```

```
// capacity()
```

```

        System.out.println("Capacity of Vector: " + v.capacity());

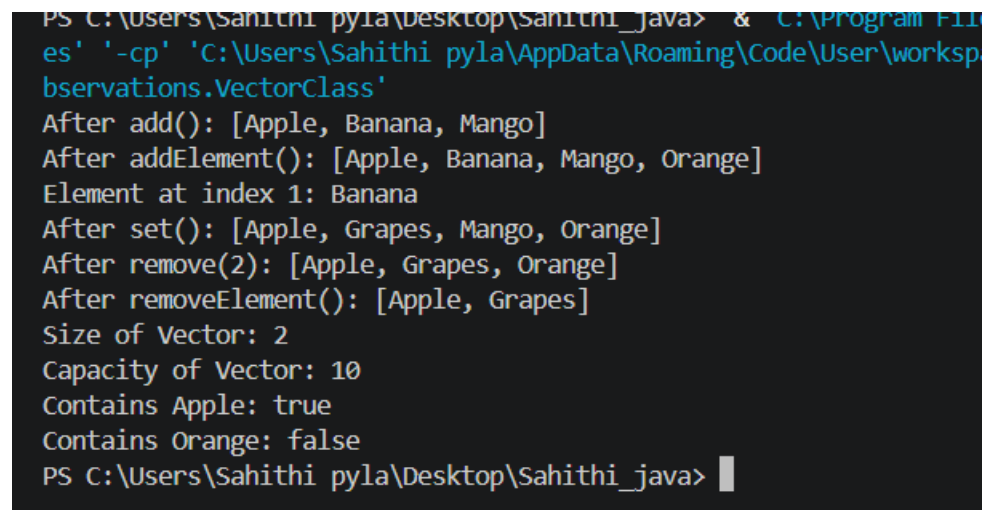
        // contains()

        System.out.println("Contains Apple: " + v.contains("Apple"));

        System.out.println("Contains Orange: " + v.contains("Orange"));
    }
}

```

Output:



```

PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java> & "C:\Program Files\Java\jdk-11.0.2\bin\java.exe" -cp "C:\Users\Sahithi_pyla\AppData\Roaming\Code\User\workspace\observations.VectorClass"
After add(): [Apple, Banana, Mango]
After addElement(): [Apple, Banana, Mango, Orange]
Element at index 1: Banana
After set(): [Apple, Grapes, Mango, Orange]
After remove(2): [Apple, Grapes, Orange]
After removeElement(): [Apple, Grapes]
Size of Vector: 2
Capacity of Vector: 10
Contains Apple: true
Contains Orange: false
PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java>

```

Result:

Thus, the Java program successfully demonstrates the commonly used methods of the Vector class.

For a practical record, you can also add a **flowchart** below the algorithm.